

The dependence of asteroid rotation on composition

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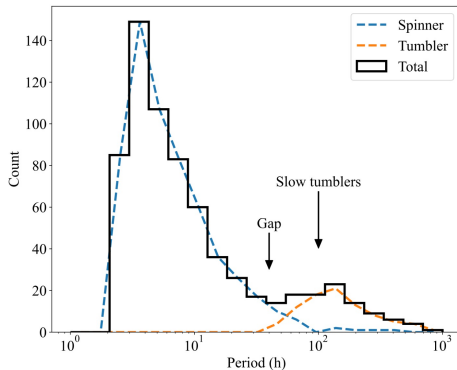
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Why study asteroid rotation on a population level?

- Asteroid spin states are shaped by collisions, YORP, and internal damping.
- Zhou et al. (2025) identifies a valley in period space a deficit between faster and slower rotators.

Rotation may be a population-scale probe of material properties.



Adapted from Zhou et al. (2025)

Science question: does composition shift the valley?

If asteroid interiors differ by composition, then the rotational valley should shift.

S-complex

Silicate-rich
ordinary-chondrite-like

⇒ stronger / less dissipative

C-complex

Carbonaceous / primitive
often more porous

⇒ weaker / more dissipative

Hypothesis: the valley location is offset between S- and C-complex asteroids

Data: building clean S- and C-complex samples

MP3C

Physical properties
diameter, albedo,
rotation period



Best values

One diameter
one period
one spectral class
per asteroid



Clean samples

S-complex: 6049
C-complex: 2947

- Spectral classes were selected using a score based on method, wavelength coverage, taxonomy scheme, and probability.
- Objects were filtered using period quality and visible albedo to reduce contamination between complexes.

A population with composition, size, spin, and albedo.

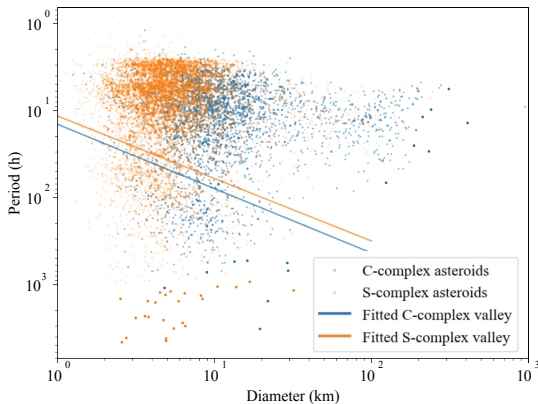
First result: the 5–10 km sample shows an offset

- Consider asteroids with $5 < D < 10$ km.
- Both complexes show a local minimum in period.
- The C-complex minimum occurs at longer periods than the S-complex minimum.

`photos/period_histogram.pdf`

Method: fitting the rotational valley

The valley is fitted as a boundary in log period–log diameter space.



Fitting approach

Semi-supervised **Support Vector Machine** method from Zhou et al. (2025), applied separately to each spectral complex.

Functional form

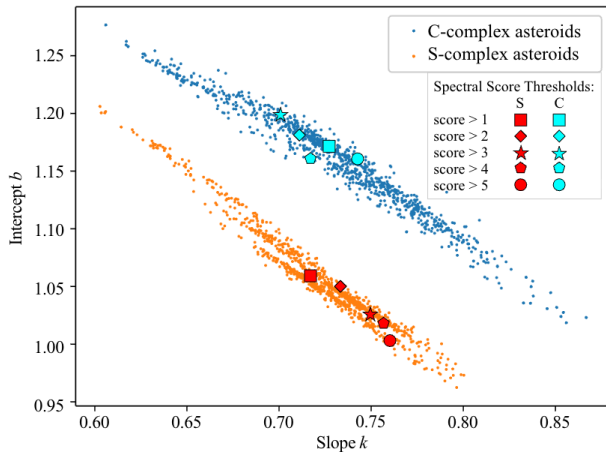
$$P^*(D) = b \left(\frac{D}{1 \text{ km}} \right)^k$$

Robustness

- The effect remains after uncertainty and spectral-score tests.

$$P_C^*(D) > P_S^*(D)$$

The rotational valley is composition dependent.



Interpretation: composition may affect spin damping

Observed result

- The C-complex valley occurs at longer periods than the S-complex valley.
- This suggests different damping behaviour between the two populations.

Physical interpretation

- Collisions can excite tumbling rotation.
- Internal friction damps this back toward principal-axis rotation.
- More dissipative bodies damp excited spin states more efficiently.

Possible physical picture

C-complex

- more porous
- weaker aggregates
- more dissipative

S-complex

- more coherent
- higher rigidity
- less dissipative

Conclusions

- We extended MP3C with a best spectral class and combined taxonomy with periods and diameters.
- S- and C-complex asteroids both show a rotational valley.
- The C-complex valley occurs at longer periods than the S-complex valley.
- This provides quantitative evidence that asteroid spin distributions depend on composition.
- Future survey data should allow this test to move from broad complexes to subclasses and families.

Take-home message

**Asteroid rotation (probably)
depends on composition.**



Please read the paper for more info!

Supplement: sample definition and contamination control

Taxonomic complexes

S-complex S, Sa, Sq, Sr, Sv

C-complex B, C, Cb, Cg, Cgh, Ch

Final samples

S-complex: **6049**

C-complex: **2947**

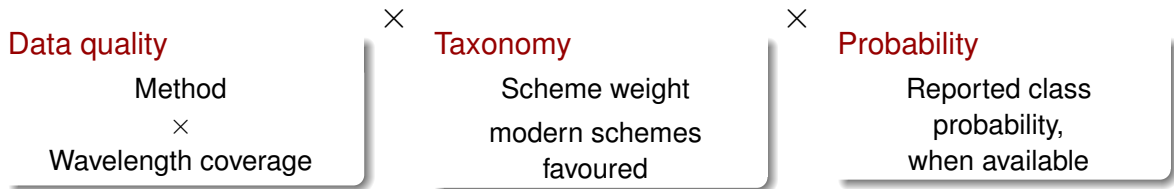
Filtering choices

- Required diameter, albedo, period, and best spectral class.
- Period quality: at least 2—.
- C-complex: $p_V \leq 0.12$.
- S-complex: $p_V > 0.12$.

**Goal: reduce contamination
between broad compositional
groups.**

Supplement: assigning one best spectral class per asteroid

For each asteroid, all available classifications were combined into a single representative class.



Selection rule

Scores are summed by class, and the class with the highest total score becomes the best spectral class.

Supplement: why show the 5–10 km slice?

- The 5–10 km range contains enough objects in both complexes.
- YORP torques are still effective in this size regime.
- The valley is visible as a local minimum in the period histogram.

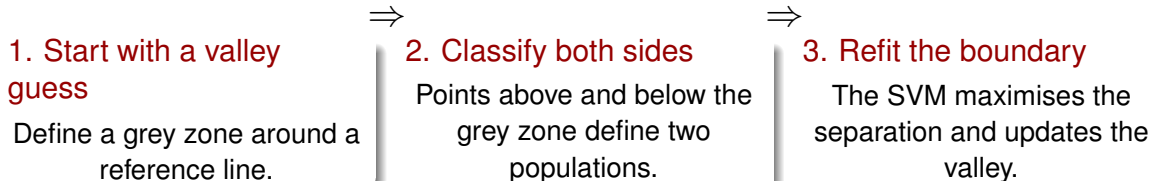
Why not all sizes in the histogram?

- Smaller bodies are less well sampled.
- Larger bodies experience weaker YORP torques.
- The full fit is still performed in $P - D$ space.

The histogram is a simple visual check; the main result comes from the full valley fit.

Supplement: what the semi-supervised SVM is fitting

The method finds the low-density boundary between faster and slower rotators in log period–log diameter space.



Fitted relation

$$\log P = k \log D + b \quad \Longleftrightarrow \quad P^*(D) = b \left(\frac{D}{1 \text{ km}} \right)^k$$

Supplement: uncertainty propagation in the valley fit

Diameter uncertainty test (bootstrapping)

- Generate 1000 synthetic datasets.
- Perturb each diameter using its quoted uncertainty.
- Refit the valley for each synthetic sample.

Main conclusion

- The offset between complexes remains.
- The slope difference is not the key claim.
- The robust result is the systematic displacement of the valley.

$$P_C^*(D) > P_S^*(D)$$

The composition-dependent offset survives the uncertainty test.

Supplement: robustness to spectral-class quality

- The nominal sample uses all objects with a valid best class.
- The test is repeated using stricter spectral-score thresholds.
- Higher thresholds keep only better-supported classifications.

Thresholds tested

score > 1 , > 2 , > 3 , > 4 , > 5

Interpretation

The result is not driven by low-score or poorly supported spectral classifications.

The valley offset remains within the uncertainty cloud.

Supplement: physical meaning of μQ

What is μ ?

- Effective rigidity.
- Measures resistance to shear deformation.
- Higher μ : more rigid body.

What is Q ?

- Quality factor for dissipation.
- Lower Q : energy is lost more efficiently.

Physical interpretation

- Large μQ : weak dissipation, slow damping.
- Small μQ : strong dissipation, rapid damping.

$$(\mu Q)_C < (\mu Q)_S$$

C-complex bodies are inferred to be more dissipative.

Supplement: why only S- and C-complex?

Why S and C?

- Broad, well-populated complexes.
- Clearer compositional interpretation.
- Linked to ordinary and carbonaceous chondrite-like material.

Why not X, D, T, or V?

- X-complex mixes several compositions.
- Rarer classes have smaller samples.
- Subclass-level tests need more data.

Future surveys should make subclass and family-level tests possible.

Supplement: future extensions

Next science steps

- Move from broad complexes to subclasses.
- Test individual asteroid families.
- Include rarer classes when samples improve.

Why this matters

- Links spin evolution to internal structure.
- Helps interpret binary and tumbling populations.
- Relevant for NEO and planetary-defence studies.

Better survey data will turn this into a stronger material-property diagnostic.